ADEPT · TEAM PECEL LELE

SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by XMUM AI Club
Submission Deadline: 1st November 2026

1. Team & Project Overview

Field	Response
Team Name	Pecel Lele
Project / Solution Name	Adept
Name (Leader)	David Kurniawan
Email & Phone (Leader)	*To be completed by team leader*
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://adept-my.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/pecel-lelele/adept

2. Architecture & Technology Stack

2.1 System Architecture Overview [Maximum 200 words]

Next.js serves a multilingual public interface and proxies requests to a private FastAPI service. Signed accounts and membership checks protect workspaces. Replit's managed PostgreSQL stores profiles, reference catalogues, snapshots, programmes and progress; local development and CI use PostgreSQL with pgvector. The launcher checks API and database readiness; it does not migrate or seed on startup.

A daily GitHub Actions schedule sends an authenticated trigger to the deployed application. Its shared worker uses its own database connection to retrieve public research, job feeds and course facts, run bounded checks, call the dedicated RunPod MiniLM endpoint for research embeddings, and atomically publish validated, dated results. Page reads reuse stored results. Historical fixtures and source outages are labelled in the interface. Deterministic matching and pathways combine confirmed skills with catalogue records and scenario estimates. Optional Gemini embeddings suggest unfamiliar skills, subject to user confirmation. Optional GLM-5.3 with Gemini failover selects checked Adi answer IDs; only server-authored text is displayed.

The original GitHub runner could not reach Replit's private database host, so refresh moved into the application; production activation is confirmed by the first successful scheduled publication.

Block flow: public sources → application-side bounded worker (with RunPod encoder) → validated PostgreSQL snapshot → private FastAPI → Next.js; signed workspaces → deterministic role, pathway and SDG views.

2.2 Languages, Frameworks & Libraries

Name & Version	Purpose in Project	License
Python 3.11; Node.js 24; pnpm 10.29.3	API and pipeline runtime; web runtime and package management	Python PSF-2.0; Node.js MIT; pnpm MIT
Next.js 16.4.2; React and React DOM 19.3.1; TypeScript 5.9.3	Public web interface, server-side proxy and static typing	Next.js and React MIT; TypeScript Apache-2.0
Tailwind CSS 4.1.18 (with its PostCSS plugin 4.1.18); class-variance-authority 0.7.1; clsx 2.1.1; tailwind-merge 3.3.1	Styling and UI component variants	MIT, except class-variance-authority Apache-2.0
next-intl 4.14.9; Recharts 2.15.4; Three.js 0.180.0; lucide-react 1.61.0	English, Malay and Chinese interface; chart components; 3D Adi mascot; icons	next-intl, Recharts and Three.js MIT; lucide-react ISC
Inter variable font package 5.3.0 (Fontsource)	Installed font package; the current interface renders with system fonts	SIL Open Font License 1.1
openapi-fetch 0.14.1; openapi-typescript 7.9.1	Typed API client generated from the API contract	MIT
FastAPI 0.161.1; Pydantic 2.14.1; pydantic-settings 2.16.1; Uvicorn 0.54.0	API framework, strict validation, configuration and ASGI server	FastAPI and Pydantic MIT; Uvicorn BSD-3-Clause
psycopg 3.3.6 (binary) and psycopg-pool 3.3.3; PostgreSQL 16 with pgvector	Relational and vector storage, connection pooling	psycopg LGPL-3.0; PostgreSQL and pgvector PostgreSQL License
PyMuPDF 1.28.2; python-multipart 0.0.32; HTTPX 0.28.1	CV text extraction (PDF), file uploads, outbound requests	PyMuPDF AGPL-3.0 or Artifex commercial licence; python-multipart Apache-2.0; HTTPX BSD-3-Clause
RunPod SDK 1.12.0; sentence-transformers 6.2.0; Transformers 5.20.2; PyTorch 2.14.1	Hosted MiniLM embedding worker (separate container image)	RunPod SDK MIT; sentence-transformers and Transformers Apache-2.0; PyTorch BSD-3-Clause
pytest 9.1.1; Ruff 0.17.2; ESLint 9.39.5; Playwright 1.64.1; Prettier 3.9.10; PyYAML 6.0.3; actionlint 1.7.7; uv	Tests, linting, formatting, browser journeys, workflow checks and Python environments	MIT, except Playwright Apache-2.0 and uv MIT or Apache-2.0

Versions are those pinned in the project's lock files and installed in the development environment on 1 November 2026; they are not an attested inventory of the deployed container. Licences were read from installed package metadata where available.

Docker Compose is used only as local development tooling for the database, not as a cloud host. PyMuPDF's AGPL-3.0 terms require review, or a commercial licence, before commercial distribution; Adept's own Apache-2.0 licence does not override third-party terms.

3. APIs & Cloud Services

3.1 External APIs

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
arXiv API (Cornell University), cs.AI listings	Public research observations for trend signals; the October 2024 replay stays labelled historical	Free public API, subject to arXiv terms and rate limits

UN Statistics Division SDG API	Official goal and target definitions; Adept's links are team alignments, not UN endorsement	Free public API
Microsoft Learn Catalog API; Coursera course API; course provider web pages	Course availability, duration and fees, with retrieval dates	Free public endpoints; provider terms apply, no resale licence
Bank Negara Malaysia Open API	Exchange rates for course prices quoted in foreign currency	Free public API
Greenhouse, Lever and Ashby public job-board APIs	Documented employer job feeds, filtered to Malaysian locations for automatic aggregate counts	Free public endpoints; reuse rights are not blanket-granted
Himalayas, Jobicy, Remote OK, Remotive	Dated remote job postings in the automatic jobs path	Free public access under each provider's terms
SmartRecruiters, Workday and Jibe career sites	Legacy connectors in the stand-alone command-line tool only; not used by the automatic refresh	Free public pages under each provider's terms
Careerjet, Jooble, Web3.career	Optional keyed job connectors; skipped when no key is configured and not required by the automatic path	Keyed access; activation and pricing not established
OpenDOSM monthly Labour Force Survey	Malaysian national labour context only, not role-level vacancy demand	Free public data, CC BY 4.0
RunPod Serverless (separate MiniLM endpoint and GLM-5.3 Adi endpoint)	Batch research embeddings; optional Adi answer selection	Sponsor-provided credits, then pay per use
Google Gemini API (3.5 Flash Lite, 3.6 Flash, 3.1 Flash Lite; gemini-embedding-001)	Optional Adi failover and unfamiliar-skill suggestions; optional trend matcher after an equivalence test	Free or paid tier depending on the account; off by default
GitHub REST API (workflow dispatch)	Optional administrator fallback that starts the trend-refresh workflow	Free with a scoped token

The **US BLS 2023–2033 projections** are a committed, dated extract, not an API called per user. MiniLM model weights are downloaded from Hugging Face when the worker image is built; the model runs on RunPod. Public reachability does not grant unlimited redistribution rights. No API keys are listed in this document.

3.2 Cloud Platforms & Deployment

Service & Provider	Purpose in Project	Plan / Region
Replit Autoscale	Public web server and private loopback API for the live prototype	Sponsor-provided Replit Core subscription; Asia region, 2 vCPU / 4 GiB at the time of the proof-of-use capture
Replit managed PostgreSQL	Separate development and production relational databases	Usage-billed under the Replit account; backup policy not yet tested
RunPod Serverless	MiniLM embedding worker and optional GLM-5.3 Adi endpoint; an earlier GPU Pod test was a separate one-off workload	Pay-per-use Serverless funded by US$25 sponsor credits; region not recorded
GitHub repository, GitHub Actions and GitHub Container Registry	Source code, CI, daily refresh trigger, worker image registry and manual fallback refresh	Public repository on GitHub-hosted runners (free tier)
Docker Compose (local)	PostgreSQL with pgvector for development and tests only	Local machine, not deployment

The Replit proof-of-use screenshots show hosting and database screens and **US$6.38 account spend in one displayed billing window**, not an Adept-only running cost. The RunPod proof-of-use screenshots show an earlier 110-skill, 0.354-second Pod vectorisation test and the Serverless set-up; they do not show that today's 142-skill catalogue was refreshed in production. Neither represents a per-learner cost.

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
all-MiniLM-L6-v2 (Sentence Transformers), pinned to a fixed Hugging Face revision	Hugging Face weights served by the team's RunPod Serverless worker	Pre-trained, not fine-tuned; 384-dimensional embeddings of public papers and catalogue skills	Apache-2.0
GLM-5.3 (Z.ai)	Hosted through RunPod for optional Adi answer selection	Pre-trained hosted inference; **selects answer-block IDs only**, never displayed prose	Z.ai model terms and RunPod terms
Gemini 3.5 Flash Lite; Gemini 3.6 Flash; Gemini 3.1 Flash Lite	Google Gemini API, optional failover chain for Adi	Pre-trained hosted inference; off by default	Gemini API Terms of Service
gemini-embedding-001	Google Gemini API, optional embedding path	Pre-trained; 768-dimensional unfamiliar-skill suggestions and an optional trend matcher; no fine-tuning	Gemini API Terms of Service

Trend method: arXiv cs.AI submissions and a dated US BLS extract → cleaning and bounded sampling → sentence embeddings and mention counts → heuristic research velocity and Moderate/Fast scenario scoring → coverage, date and ID validation → stored, dated snapshot. By default the trend matcher uses MiniLM and switches to Gemini embeddings only after a passing method-equivalence report; an operator can force Gemini manually, in which case the series is not equivalence-validated and is stored separately. Research attention is **not** observed Malaysian labour demand, and scenario bands are **not** calibrated confidence intervals. The default runtime uses no language model.

4.2 Datasets

Counts are **rows in the source catalogue files on 1 November 2026**, not a deployed-database count: **142 skills; 48 roles, of which 33 are target roles; 104 courses**. Role and skill links are reviewed team curations. Earlier proof-of-use screenshots and a 20 October 2026 synthetic impact fixture show smaller, earlier catalogue sizes. No team-trained model and no real CV dataset is used.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Curated roles and skills, role requirements and O*NET 31.0 crosswalk	Team curation; O*NET Resource Center (U.S. Department of Labor)	Team-curated, public-derived	O*NET CC BY 4.0 with stated exceptions; edited requirement levels are Adept estimates	N

arXiv cs.AI paper metadata and dated mention counts	arXiv (Cornell University)	Public metadata, sampled extracts	arXiv metadata CC0 1.0; individual papers keep their own licences	Y (author names in public metadata; no learner records)
US occupational projections 2023–2033	U.S. Bureau of Labor Statistics	Public, derived extract	U.S. Government public domain; US-only baseline, not a Malaysian forecast	N
Historical October 2024 trend replay and synthetic trend fallback	Team-created from dated arXiv metadata, plus a synthetic fallback	Public-derived and synthetic	Team-created; values labelled historical or fixture	N
UN SDG goals (17) and targets (169); team role (88) and skill (167) alignment rows	UN SDG API (UN Statistics Division); team curation	Public definitions and team-curated mappings	UN terms of use; mappings are team hypotheses, not official approval	N
104 curated courses and skill prerequisites	Microsoft Learn; Coursera; other named course providers	Public facts and team curation	Content, fees and rights remain with providers; HRD claimability must be checked per course	N
1,175 job-posting rows; certification list; exchange-rate extracts	Greenhouse, Lever and Ashby job boards; certification issuers; Bank Negara Malaysia	Public, curated	Job-board and issuer terms apply; older postings may have closed	N for learners; postings name employers
Monthly Labour Force Survey; MyCOL 2024/2025 critical occupations	OpenDOSM (Department of Statistics Malaysia); TalentCorp MyCOL	Public aggregates and reviewed crosswalk	OpenDOSM CC BY 4.0; TalentCorp terms; the MyCOL badge does not affect ranking	N
Synthetic demonstration profiles, employers, CVs and test fixtures	Team-created	Synthetic	Team-created	N

4.3 AI-Assisted Development Tools

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Anthropic Claude Code (model versions varied over the project)	Code generation, debugging, documentation and submission drafting	Major
OpenAI Codex (model versions varied over the project)	Debugging, implementation review and code and document corrections	Major
Replit Agent	Replit deployment configuration, publishing and repository exploration	Minor
Google Antigravity agent	Repository exploration	Minor
Greptile	Automated pull-request review comments	Moderate
ElevenLabs (synthesis model/ version not recorded)	Presentation/demo voice-over using a stock voice; the team confirmed historical use on 30 October 2026. Not an Adept runtime integration.	To be confirmed by the team

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output: [Maximum 100 words]

The team set Adept's product direction and reviewed changes through pull requests before merging. Claude Code produced much of the source and document drafts; Codex helped debug and review; Replit Agent, Antigravity and Greptile supported deployment set-up, inspection and pull-request feedback. Team members remain responsible for all accepted code and claims. Changes pass automated lint, type, contract, attribution and fairness tests and are checked against source records and dates; passing tests do not verify a deployed build or customer benefit. Model output never authorises invented catalogue entries or personnel decisions.

5. Sponsor-Provided Resources

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

The team used a sponsor-provided Replit Core subscription and US$25 of RunPod credits. Sponsorship is not permanent free production hosting.

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Core subscription, Autoscale deployment, managed PostgreSQL	Public prototype, private API and persistent application records; Replit Agent use is declared in Section 4.3.
RunPod	US$25 credits; GPU Pod test; Serverless MiniLM and GLM-5.3 endpoints	Research embeddings and optional Adi inference; the one-off Pod test validated batch encoding.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly: [Maximum 100 words]

Replit is the prototype's current web, API and managed-PostgreSQL host; RunPod provides MiniLM encoding and optional Adi inference. Neither sponsor's credit is a viable permanent business model. Replacing Replit would require migrating the same schema, signed-session boundaries and backups; replacing MiniLM requires a method-equivalence check before changing any scored series. The learner's normal matching and plan reads are deterministic and do not require per-page model calls. Provider outages preserve dated data or show unavailable states, so the core journey continues without either model service.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy [Maximum 150 words]

The demo uses synthetic identities and CVs, not real applicants. Signed accounts can store names, emails, password hashes, confirmed skills, preferences, memberships, plans, progress and evidence links in PostgreSQL. CV bytes are parsed in memory and not retained. Members explicitly opt in to share skills and progress; personal plans and assistant conversations stay separate. Authentication, server-side membership and manager checks, upload limits, bounded network access and HTTP-only signed cookies mitigate risks.

Optional Gemini suggestions receive short contact-redacted skill phrases. Optional Adi providers receive filtered context and server-authored candidate blocks; redaction cannot guarantee that no personal information leaves the service. MiniLM receives public research and catalogue text. Account export and deletion exist, but automatic retention, tested backup erasure and contractual real-user processing arrangements are **not** established. Adept does not claim PDPA certification, and no real CVs are used in the demo.

6.2 Known Limitations & Risks [Maximum 150 words]

arXiv research share and a US labour baseline cannot measure Malaysian occupation demand. Historical

2024 replay is not fresh evidence; an online check can leave observations unchanged. Scenarios and skill exposure are heuristics, not calibrated probabilities. The catalogue covers selected roles and courses; fees, openings, prerequisites and HRD claimability can change. A completion click is not verified competence or a regulated qualification. SDG links are draft enabling hypotheses, not measured UN outcomes.

Source dates, estimate and fallback labels, ID validation, permissions, bounded workers, stale states and human confirmation mitigate risks; they do not prove universal fairness. A successful model call does not publish data. A local test, public HTTP 200 or sponsor screenshot cannot establish the deployed revision, current observations, vendor licences or customer outcomes. Each requires separate verification before commercial use.

7. Compliance & Readiness Checklist

☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☐ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☑ Any use of official SDC sponsor resources has been declared in Section 5.

☐ The prototype described here is accessible to judges through the links provided in Section 1.

☐ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature: _____

Date: _____